

 AI4ICPS IIT Kharagpur	IIT KHARAGPUR AI4ICPS I HUB FOUNDATION Hands-on Approach to Advanced AI for Real-World Applications (HAAI)++ - Cohort 2 Programming Assignment 3: In-context Learning in LLMs
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Due date: Sunday 9th August 2026, EOD – IST

Important Instructions about Programming Assignments

1. Programming assignments will be evaluated automatically. **Do not** change the skeleton/template code provided to you.
2. Write your code **only in the designated places** in the skeleton/template code and process the input data provided to you in the designated variables. **Do not alter** the input-output structure in the skeleton code.
3. **Do not import** any additional libraries. **Do not use any additional files** for the processing (other than those mentioned in the skeleton code).
4. Failure to comply with these instructions may lead to you getting **zero marks** for the assignment, even if the solution is largely correct.
5. Remember that your output will be evaluated against test cases, therefore any deviation from the test cases will be considered incorrect during evaluation.

Objective: To use Flan-T5-XL LLM to be able to answer a question in YES or NO using the provided context as in-context learning.

Problem: Consider the two inputs: Context, Question. Here, context is a passage containing some information about a person, place or anything. The question is specifically designed to answer something from the context. Therefore, the LLM should have access to the context in order to answer the question.

Instructions: The program accepts two texts provided as a command line input. The two inputs represent the context and the question respectively. Consider that the question can be always answered in YES or NO; the context is within 30 words. The question output is deterministic i.e., its output should be either YES or NO. You are required to use logits to extract the output. Output should be in upper-case. There should be no additional output including any warning messages in the terminal. Design the code to run in CPU environment. It is recommended to use terminal to avoid inconsistency in output formatting.

Execution Syntax: `python template.py <CONTEXT> <QUESTION>`

The following examples are given for your reference:

Terminal Input: 'John started watered his plants every morning this week.' 'Did John water his plants yesterday morning?'
Terminal Output: YES

Terminal Input: 'John started watered his plants every morning this week.' 'Did John water his plants last month?'
Terminal Output: NO

****Additional examples are provided in the template.py file for your reference.**

NOTE: *You should create some similar examples on your own to test the correctness of your pipeline.*